Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad, Natore

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Lesson Plan

Program:	Computer Science and Engineering (CSE)
Course Code & Title:	CSE-2105 Database Systems
Semester:	Summer-2025
Level:	2nd Year 1st Semester
Credit Hour:	3.00 Hr.
Name & Designation of Teacher:	**Md. Omar Faruq** Assistant Professor, Dept. of CSE **Md. Rafiqul Islam (Rafiq)** Lecturer, Dept. of CSE
Class Hours:	Sunday: 10:15 AM-11:10 AM Tuesday: 10:15 AM-11:10 AM Wednesday: 08:15 AM-09:10 AM
Consultation Hours:	Wednesday:11:40AM -02:55PM
	Monday: 08:15 AM-11:10 AM
E-mail:	omarfarukcse10@gmail.com (Md. Omar Faruq) rafique.cseduet@gmail.com(Md. Rafiqul Islam (Rafiq))
Mobile:	+8801708507424 (Md. Omar Faruq) +8801734249264 (Md. Rafiqul Islam (Rafiq))
Rationale:	This course is designed to introduce the basic concepts of database, learn the foundations of database systems, focusing on basics such as the relational algebra and data model, schema normalization, query optimization, and database designing.
Pre-requisite(if any):	None

Objectives:

Upon completion the course, the student will be able to:

1. To understand the basic concepts and appreciate the applications of database systems, identify the data models and DBMS structure.
2. To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.
3. To demonstrate the basics of SQL and construct queries using SQL and introduction to the relational database theory and be able to write relational algebra expressions for queries.
4. To develop an understanding of essential DBMS concepts such as: database security, integrity, and concurrency.

Course Content:

Mid Term

Concepts of Database Systems: Files and Databases, Database Management Systems; Transaction management, Structure of a DBMS, Applications.

Entity-Relationship Concepts: Entity types, Entity set, Attribute and key, Relationships, Relation types, Entity relationship, ER modeling, ER diagrams.
Relational Model: Structure of relational databases, Relational algebra, and Relational algebra operations.
SQL: Data Definition Language, Data Manipulation Language, Basics of SQL.

Semester Final

Concepts of Database Systems: Files and Databases, Database Management Systems; Transaction management, Structure of a DBMS, Applications.
Entity-Relationship Concepts: Entity types, Entity set, Attribute and key, Relationships, Relation types, Entity relationship, ER modeling, ER diagrams, Database design using ER diagrams, Enhanced Entity- Relationship (EER) model.
Normalization: Normal forms, Normalized Relations and Database performance; De-normalization.
Relational Model: Structure of relational databases, Relational algebra, Relational algebra operations, Modification of the database, Introduction to views, Pitfalls in relational database design.
SQL: Data Definition Language, Data Manipulation Language, Basics of SQL, Query designing in SQL using aggregate functions and nested queries, Embedded SQL, Triggers, Procedures; Indexes; Declarative Constrains and Database Triggers.
Concurrency Control: Lock based protocols, Timestamp based protocols, and Validation based protocols, Deadlock.
Recovery System: Failure classification, Storage structure, Recovery and atomicity, Log based recovery, Recovery with concurrent transactions, Advanced recovery techniques, RAID model.
Advanced Database Management Systems: No SQL Systems, distributed systems, object-oriented System, Temporal, Database Security, Data Warehousing and Data Mining, Database Administration and Tuning.

Text Book(s):

1. DatabaseSystemConcepts,Mcgraw-Hill-A.Silberschatz,7th Edition,ISBN- 9780078022159
2. DatabaseManagementSystem,McGraw-HillHigherEducation-R.Ramakrishnan, JohannesGehrke,3rd Edition,ISBN-0-07-24-6563-8

Reference Book(s):

1. PrinciplesofDatabaseManagement,Prentice-hall-JamesMartin
2. DatabaseManagementsystems,Prentice-HallPublication-Ullman
3. Oracle8iaBeginnersGuide,McGrawHill-Abey

***Additional reading material:** Power Point slides, Docx & PDF provided by the Teacher.

Course Assessment Methods & their Weights:

Assessment Method	**(%)**
Class Test/Assignment	15
Class performance	05
Mid-Term Assessment	20
Semester Final Exam	60
Total:	**100**

Minimum Attendance: 70%classattendanceis mandatory for a student in order to appear at the final examination.

Student's responsibilities:

- **a.** Students must come to the class prepared for the course material covered in the previous class.
- **b.** They must submit their assignments on time.
- **c.** No late or partial assignments will be acceptable.
- **d.** There will be no make-up Class Test.

Grading Policy:

Numerical Score	**Letter Grade**		**Grade points**
80% and above	A+	(A Plus)	4.00
75% to less than 80%	A	(A Regular)	3.75
70% to less than 75%	A-	(A Minus)	3.50
65% to less than 70%	B+	(B Plus)	3.25
60% to less than 65%	B	(B Regular)	3.00
55% to less than 60%	B-	(B Minus)	2.75
50% to less than 55%	C+	(C Plus)	2.50
45% to less than 50%	C	(C Regular)	2.25
40% to less than 45%	D	(D Pass)	2.00
Less than 40%	F	Fail	0.00
Incomplete	I	-	-
Withdrawal	W	-	-
Project/Thesis Continuation	X	-	-

Detailed Course Plan and Mapping (According to BAETE Accreditation Manual- II)

i. Mapping CO-PO, Bloom's Level, Knowledge and Attitude Profile (WK), Complex Engineering Problem Solving (WP) and Complex Engineering Activities (EA)

Course No & Title: CSE-2105 (Database Systems)

COs	CO Statements Upon completion of the course, the students will be able to)	PO	WK	WP	EA	Bloom's Level	Delivery methods and activities	Assessment Tools
CO1	Understand the basic concepts of database systems, appreciate their applications, and identify data models and DBMS structures for solving complex engineering problems.	1	1	2	1	C2, C3	Lectures, Text Book, PPT, PDF White Board Writing	Class Participation/ Presentation, Class Performance, Homework/ Assignment, Class Test, Semester Final Examination (Written)
CO2	Apply fundamental knowledge of database systems, including modeling, designing a DBMS, to develop and construct simple database solutions for engineering applications.	1	3	2		C2, C3		
CO3	Identify, formulate, and analyze problems related to relational databases by demonstrating the basics of SQL, constructing queries, and applying relational algebra expressions using MySQL/Oracle, based on fundamental principles of mathematics and engineering sciences.	2	4	1		C2, C3		
CO4	Develop a comprehensive understanding of key DBMS concepts, including transactions, serializability, database security, integrity, and concurrency control.	2	3	2		C2, C3		

Legends: CO-Course Outcomes PO-Program Outcomes, KP-Knowledge Profile
Bloom's Level: Cognitive Domain (C1-C6), Psychomotor Domain (P1-P7), and Affective Domain (A1-A5)

Cos	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	H											
CO2	M											
CO3		H										
CO4		L										

Legends: High (H)=3, Medium(M)=2 and Low(L)=1

Assessment of CO-PO, Bloom's Level

Teaching-learning and Assessment Strategy:
Class Participation/Presentation, Class Performance, Homework/Assignment, Class Test, Semester Final Examination (Written)

Assessment Methods	Description	Grading	CO	Blooms Taxonomy	
Continuous Assessment (30%)	Class Test/Assignment	15%	CO1 CO2 CO3	C2,C3	
	Class Performance	5%	CO4	C2	
Mid Term Assessment (20%)	Examination/Project	20%	CO1 CO3	C2, C3	Answer two questions including Q.1
Final Examination (60%)	Section A	30%	CO1 CO2 CO3 CO4	C2, C3	Question 1 and 2 from Section A and Question 5 and 6 from Section B will satisfy specified CO-POs, BLs, KPs and WPs
	Section B	30%			
Total Marks		100%			

Teaching and Learning Strategies (TLS)	
TLS1	Delivery lectures using multimedia of different topics
TLS2	Active and fruitful discussion in class regarding efficient solving of the logical, mathematical and complex engineering problems
TLS3	Group discussion and presentation regarding diverse problems and corresponding lectures
TLS4	Evaluation of class performance to reach each student in a class for every topic.

Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) represent the knowledge, skills and attitudes the students should have at the end of a four years engineering program. CSE program of BAUET has 12 Program Outcomes. They are briefly described in the following table.

PO	Category	Description
PO1	Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems

PO2	Problem Analysis	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
PO3	Design creative solutions	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5).
PO4	Investigation	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5	Modern Tool Usage	Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
PO6	The Engineer: Society, Economy, Environment and Sustainability	When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).
PO7	Ethics	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).
PO8	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9).
PO9	Communication	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences
PO10	Project Management	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team and to manage projects and in multidisciplinary environments
PO11	Life Long Learning	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8).
PO12	Entrepreneurship	Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.

Knowledge and Attribute Profile (KPs)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline

WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Range of Complex Engineering Problem Solving (Need High Taxonomy Level)

Attribute: Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7

WP1	Depth of Knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
WP2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
WP3	Depth of analysis required	Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
WP4	Familiarity of issues	Involve infrequently encountered issues or novel problems
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice for professional engineering
WP6	Extent of stakeholder involvement and conflicting requirements	Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
WP7	Interdependence	Address high level problems including many components or sub-problems that may require a systems approach

Range of Complex Engineering Activities (Project based)

Attributes: Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:

EA1	Range of resources	Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
EA2	Level of interaction	Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues

EA3	Innovation	Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
EA4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
EA5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches

Bloom's Taxonomy Domain:

Cognitive Domain		**Psychomotor Domain**		**Affective Domain**	
C1	Remembering	**P1**	Perception	**A1**	Receive
C2	Understanding	**P2**	Set	**A2**	Respond
C3	Applying	**P3**	Guided Response	**A3**	Value
C4	Analyzing	**P4**	Mechanism	**A4**	Organize
C5	Evaluating	**P5**	Complex Overt Response	**A5**	Internalize
C6	Creating/Designing	**P6**	Adaption		

Detail Lecture Plan

Module-1:

Lecture 1: Structure of Relational Databases
Course Outcome: CO1
Learning Outcomes:

1. Understand the architecture and schema of relational databases.
2. Identify keys and constraints used in relational models.
   Topics Covered:

- Tables, attributes, tuples
- Schema and instance
- Primary key, foreign key, candidate key, super key
  Review Questions:
- What are the differences between schema and instance?
- Define and give examples of candidate and foreign keys.

Lecture 2: Relational Algebra - Part I
Course Outcome: CO3
Learning Outcomes:

1. Apply basic relational algebra operations.
2. Construct expressions using select, project, and union.
   Topics Covered:

- Select (σ), Project (π), Union ($\cup$), Difference ($-$)
  Review Questions:
- Write relational algebra expressions for retrieving data from a student table using projection and selection.

Lecture 3: Relational Algebra - Part II
Course Outcome: CO3
Learning Outcomes:

1. Understand join operations and cross product in relational algebra.
2. Apply these in query scenarios.
   Topics Covered:

- Cartesian Product ($\times$), Join ($\bowtie$), Natural Join, Theta Join
  Review Questions:
- Demonstrate the use of different joins in relational algebra.

Lecture 4: Advanced Relational Algebra
Course Outcome: CO3
Learning Outcomes:

1. Analyze query optimization using relational algebra.
2. Apply set intersection, division, and nested queries.
   Topics Covered:

- Intersection, Division (÷), Nested operations
  Review Questions:
- Write a relational algebra expression to find sailors who reserved all boats.

Lecture 5: Normalization Concepts
Course Outcome: CO2
Learning Outcomes:

1. Explain the purpose and process of normalization.
2. Identify anomalies in un-normalized data.
   Topics Covered:

- Need for normalization, insertion, update, and deletion anomalies
  Review Questions:
- Why do anomalies occur in database design? How does normalization help?

Lecture 6: First and Second Normal Forms
Course Outcome: CO2
Learning Outcomes:

1. Apply 1NF and 2NF to given relations.
2. Eliminate partial dependencies.
   Topics Covered:

- Functional dependencies
- Conversion to 1NF and 2NF
  Review Questions:
- Normalize a table with repeating groups to 2NF.

Lecture 7: Third Normal Form and BCNF
Course Outcome: CO2
Learning Outcomes:

1. Distinguish between 3NF and BCNF.
2. Normalize complex schemas using functional dependencies.
   Topics Covered:

- 3NF, BCNF rules and examples
  Review Questions:
- Convert a 2NF table to 3NF and then to BCNF with justification.

Lecture 8: De-normalization and Performance
Course Outcome: CO2
Learning Outcomes:

1. Understand when and why to de-normalize.
2. Assess trade-offs between normalization and performance.
   Topics Covered:

- De-normalization scenarios, performance tuning
  Review Questions:
- Give examples of situations where de-normalization is beneficial.

Lecture 9: Lock-based Concurrency Control

Course Outcome: CO4
Learning Outcomes:

1. Analyze concurrency issues and solutions.
2. Apply lock-based protocols.
   Topics Covered:

- Shared and exclusive locks, 2PL (Two-Phase Locking)
  Review Questions:
- Describe two-phase locking protocol with an example transaction schedule.

Lecture 10: Timestamp-based Protocols
Course Outcome: CO4
Learning Outcomes:

1. Explain the use of timestamps in concurrency control.
2. Compare with locking mechanisms.
   Topics Covered:

- Basic and strict timestamp ordering protocols
  Review Questions:
- Discuss how timestamp ordering ensures serializability.

Lecture 11: Validation-based Protocols and Deadlocks
Course Outcome: CO4
Learning Outcomes:

1. Understand validation phases in concurrency.
2. Detect and resolve deadlocks.
   Topics Covered:

- Validation protocols, Wait-Die and Wound-Wait, Deadlock handling
  Review Questions:
- How can deadlocks be prevented using Wound-Wait?

Lecture 12: Failure Classification and Recovery Concepts
Course Outcome: CO4
Learning Outcomes:

1. Classify types of failures in DBMS.
2. Understand recovery basics.
   Topics Covered:

- Transaction failure, system crash, storage failure
  Review Questions:
- List and explain different types of failures in a DBMS environment.

Lecture 13: Log-based Recovery and Atomicity
Course Outcome: CO4
Learning Outcomes:

1. Understand write-ahead logging and recovery.
2. Apply atomicity rules in transactions.

Topics Covered:

- Write-ahead logging (WAL), Undo/Redo

Review Questions:

- What is the significance of redo and undo logs in recovery?

Lecture 14: Concurrent Recovery Techniques
Course Outcome: CO4
Learning Outcomes:

1. Handle concurrent transaction failures.
2. Implement checkpoint-based recovery.

Topics Covered:

- Checkpoints, shadow paging, recovery with concurrent transactions

Review Questions:

- Describe the role of checkpoints in recovery systems.

Lecture 15: RAID and Storage Management
Course Outcome: CO4
Learning Outcomes:

1. Understand RAID levels and data redundancy.
2. Compare performance and fault tolerance.

Topics Covered:

- RAID levels 0–6, striping, mirroring, parity

Review Questions:

- Which RAID level provides the best balance between performance and fault tolerance?

Lecture 16: NoSQL and Non-relational Databases
Course Outcome: CO3
Learning Outcomes:

1. Differentiate between relational and NoSQL systems.
2. Understand NoSQL models and use-cases.

Topics Covered:

- Key-Value, Document, Column-Family, Graph databases

Review Questions:

- Compare relational databases with NoSQL databases in terms of scalability and flexibility.

Lecture 17: Distributed and Object-Oriented DBMS
Course Outcome: CO4
Learning Outcomes:

1. Explore features of distributed databases.
2. Describe object-oriented database components.

Topics Covered:

- Data fragmentation, replication, inheritance in OODBMS

Review Questions:

- What are the advantages of distributed databases over centralized ones?

Lecture 18: Data Warehousing, Mining, and Security
Course Outcome: CO4
Learning Outcomes:

1. Understand concepts of data warehousing and mining.
2. Discuss security measures in DBMS.

Topics Covered:

- OLAP, ETL, association rules, encryption, access control

Review Questions:

- What are the differences between OLAP and OLTP?
- Describe common security challenges in DBMS.

Module-2:

Lecture 19: Introduction to Database Systems
Course Outcome: CO1
Learning Outcomes:

1. Understand the difference between files and databases.
2. Appreciate the need for database systems in modern applications.

Topics Covered:

- Files vs. Databases
- Limitations of file-based systems
- Advantages of DBMS

Review Questions:

- Why are databases preferred over traditional file systems?
- List three advantages of using a DBMS.

Lecture 20: Structure of a DBMS
Course Outcome: CO1
Learning Outcomes:

1. Identify the components of a DBMS.
2. Understand how DBMS components interact to manage data.

Topics Covered:

- Query Processor
- Storage Manager
- Transaction Manager
- Catalog Manager

Review Questions:

- What are the main functions of a query processor?
- Describe the role of the storage manager.

Lecture 21: Database Applications and Users
Course Outcome: CO1
Learning Outcomes:

1. Recognize various types of database users and applications.
2. Understand the role of databases in real-world applications.

Topics Covered:

- Database users: DBA, application programmers, end users
- Applications: banking, education, healthcare

Review Questions:

- What are the responsibilities of a database administrator?
- Give examples of database use in real-life systems.

Lecture 22: Entity and Attribute Concepts
Course Outcome: CO2
Learning Outcomes:

1. Understand basic building blocks of ER models.
2. Distinguish between different types of attributes.

Topics Covered:

- Entities and Entity sets
- Attributes (simple, composite, derived, multivalued)

Review Questions:

- Differentiate between composite and derived attributes.
- Define entity set with an example.

Lecture 23: Keys and Relationships in ER Model
Course Outcome: CO2
Learning Outcomes:

1. Identify types of keys and their importance.
2. Understand relationship types and their role in ER modeling.

Topics Covered:

- Primary key, candidate key, super key
- Relationship types (1:1, 1:N, M:N)

Review Questions:

- Explain the difference between primary key and super key.
- Give an example of a many-to-many relationship.

Lecture 24: ER Diagram and Design Concepts
Course Outcome: CO2

Learning Outcomes:

1. Create ER diagrams based on system requirements.
2. Apply database design principles using ER modeling.

Topics Covered:

- ER diagram notation
- Mapping entities and relationships
- Constraints in ER modeling

Review Questions:

- Draw an ER diagram for a library system.
- What are cardinality constraints?

Lecture 25: Enhanced ER (EER) Modeling
Course Outcome: CO2
Learning Outcomes:

1. Understand concepts of generalization, specialization, and aggregation.
2. Extend ER models using EER features.

Topics Covered:

- Generalization, Specialization
- Aggregation

Review Questions:

- How is generalization different from specialization?
- Explain aggregation with an example.

Lecture 26: SQL Basics and DDL
Course Outcome: CO3
Learning Outcomes:

1. Write SQL statements to define database schema.
2. Apply constraints using SQL.

Topics Covered:

- CREATE, ALTER, DROP
- Primary key, foreign key, NOT NULL

Review Questions:

- Write SQL to create a table with constraints.
- How do you modify an existing table?

Lecture 27: SQL DML and Data Retrieval
Course Outcome: CO3

Learning Outcomes:

1. Retrieve and modify data using SQL commands.
2. Apply filtering and sorting in queries.

Topics Covered:

- SELECT, INSERT, UPDATE, DELETE
- WHERE clause, ORDER BY, LIKE, IN

Review Questions:

- Write a query to update salary of employees in a department.
- Explain the use of the LIKE operator.

Lecture 28: SQL Aggregate Functions and Grouping
Course Outcome: CO3
Learning Outcomes:

1. Apply aggregate functions to summarize data.
2. Group data using GROUP BY and HAVING clauses.

Topics Covered:

- COUNT, SUM, AVG, MIN, MAX
- GROUP BY, HAVING

Review Questions:

- Write a query to find the average salary by department.
- What is the purpose of the HAVING clause?

Lecture 29: Nested Queries and Joins in SQL
Course Outcome: CO3
Learning Outcomes:

1. Construct nested queries and apply different types of joins.
2. Understand subqueries and their application.

Topics Covered:

- Subqueries (single/multiple row)
- INNER JOIN, LEFT JOIN, RIGHT JOIN

Review Questions:

- Write a nested query to find employees earning more than the average salary.
- Differentiate between INNER and LEFT JOIN.

Lecture 30: SQL Constraints, Triggers, and Procedures
Course Outcome: CO3
Learning Outcomes:

1. Implement constraints and triggers to enforce data integrity.
2. Create simple stored procedures in SQL.

Topics Covered:

- CHECK, UNIQUE, DEFAULT
- TRIGGERS and stored procedures basics

Review Questions:

- What is the purpose of a database trigger?
- Write a procedure to insert a record into a table.

Lecture 31: Introduction to Indexing
Course Outcome: CO4
Learning Outcomes:

1. Understand the concept and need for indexing.
2. Compare dense and sparse indexing.

Topics Covered:

- Basics of indexing
- Dense vs. Sparse indexing

Review Questions:

- What is the difference between dense and sparse indexing?
- Why is indexing important in databases?

Lecture 32: B+ Tree Indexing
Course Outcome: CO4
Learning Outcomes:

1. Apply B+ tree insertion and deletion.
2. Understand the structure and advantages of B+ trees.

Topics Covered:

- B+ tree structure
- Insertion and deletion examples

Review Questions:

- Perform B+ tree insertion for given values.
- Why are B+ trees preferred over B-trees?

Lecture 33: Hashing and Collision Resolution
Course Outcome: CO4
Learning Outcomes:

1. Understand static and dynamic hashing.

2. Apply collision resolution techniques.

Topics Covered:

- Static vs. dynamic hashing
- Collision resolution: chaining, open addressing

Review Questions:

- Describe linear probing with an example.
- How does dynamic hashing work?

Lecture 34: Transaction Management and Properties
Course Outcome: CO4
Learning Outcomes:

1. Understand ACID properties.
2. Identify states of a transaction.

Topics Covered:

- Transaction concepts
- ACID properties
- Transaction states

Review Questions:

- What are the ACID properties?
- Explain the transaction lifecycle.

Lecture 35: Serializability and Concurrency Control
Course Outcome: CO4
Learning Outcomes:

1. Apply serializability concepts to maintain consistency.
2. Understand the need for concurrency control.

Topics Covered:

- Conflict and view serializability
- Concurrency control techniques

Review Questions:

- Define conflict serializability.
- What is the purpose of locking in databases?

Lecture 36: Database Security and Integrity
Course Outcome: CO4
Learning Outcomes:

1. Implement database security policies.

2. Maintain data integrity using constraints and triggers.

Topics Covered:

- Authentication, Authorization
- Integrity constraints, access control

Review Questions:

- Explain how access control works in databases.
- What are the types of data integrity?

Assignment: Design and Implementation of a Library Management Database System
Submission Date:
Full Marks:

Step 1: Define the Problem

Range of Conflicting Requirements (P2)

Design a database management system (Choose a **unique Database name** such as **Library_DataBase** to manage a library's operations). The system should handle entities such as Books, Members, Librarians, Loans, and Categories. Define relationships such as each member borrowing books, each book belonging to a specific category, and each loan being processed by a librarian.

Entities and Attributes:

1. **Books**
   - Primary Key: `BookID`
   - Attributes:
     - `Title` (Composite: contains `MainTitle` and `Subtitle`)
     - `Author` (Multi-valued)
     - `PublicationYear` (Stored)
     - `ISBN` (Stored)
     - `CategoryID` (Foreign Key)
2. **Members**
   - Primary Key: `MemberID`
   - Attributes:
     - `Name` (Composite: contains `FirstName` and `LastName`)
     - `MembershipDate` (Stored)
     - `Email` (Stored)
     - `PhoneNumber` (Stored)
3. **Librarians**
   - Primary Key: `LibrarianID`
   - Attributes:
     - `Name` (Composite: contains `FirstName` and `LastName`)
     - `EmployeeID` (Stored)
4. **Loans**
   - Primary Key: `LoanID`
   - Attributes:
     - `LoanDate` (Stored)
     - `ReturnDate` (Derived)
     - `MemberID` (Foreign Key)
     - `BookID` (Foreign Key)
5. **Categories**
   - Primary Key: `CategoryID`
   - Attributes:
     - `CategoryName` (Stored)

Relationships:

- **Members** borrow **Books** through **Loans**.
- **Books** are categorized under **Categories**.
- **Librarians** process **Loans**.

E-R Diagram:

Construct an Entity-Relationship diagram showing all entities, attributes, and relationships.

Relational Schema:

Formulate the relational schema for **Library_DataBase** based on the entities defined above.

Step 2: Test and Implement a Solution

Extend of Applicable Codes (P5)

1. **Create the Database Using Data Definition Language (DDL):** Write the SQL statements to create the tables for the entities defined in the relational schema.
2. **Insert Values Using Data Manipulation Language (DML):** Insert sample data into each table to illustrate how the database will function.
3. **Prove Database Usability:** Create SQL queries that demonstrate how to:
    - Retrieve a list of all books by a specific author.
    - Find all loans associated with a particular member.
    - List all books in a specific category.
    - Get the return date for a particular loan based on the loan date and current date.

Task:

- E-R diagram and relational schema of your database.
- SQL DDL statements for creating the database.
- SQL DML statements for inserting sample values for some tuples.
- SQL queries that demonstrate the ease of data retrieval and manipulation of the database.
- Relational algebra expression that demonstrate the equivalent to the manipulation of the database.

Approved by

Chairman, PSAC
Dept. of CSE, BAUET